Forecast-aware route planning for military vehicles on the Pachmarhi plateau

A from-scratch explainer of the project, the data, what was built and what it found, and what goes to the ICDCN 2027 poster track.

Updated	**13 October 2026**. Replaces the 10 October version.
Poster deadline	**15 October 2026** (2 days from this document)
Venue	ICDCN 2027, NITK Surathkal, India, 4–7 January 2027
Project folder	Documents\meteorology

What changed since the 10 October version

- **Every step of the plan is now built and tested**: vehicle profiles, terrain scores, the forecast-aware planner, the convoy experiment, a backtest over 2024–25, and the edge numbers.
- **Vehicles:** the scope is all vehicles in the file (tracked, wheeled and half-tracks), not only wheeled ones.
- **Ground firmness:** the wettest value was lowered from 25 to 15 psi (Section 4.2). Every forecasting number was recomputed.
- **Satellite data:** your GeoTIFF export arrived and replaced the picture-only maps.
- **Deliverables:** the A0 poster and the 2-page abstract are both built. Your names, logo and links are the only gaps.
- **Corrections:** ZIS-30 is a tracked tank destroyer, not a truck. The September forecasting numbers no longer apply (Section 5).
- **New: SLOW-GO in numbers.** A SLOW-GO day now also says how many vehicles one track carries and how fast the band can drive. The planner can choose routes by travel time and report trip times in minutes (Section 6, Step 8).
- **New: the whole chain on one real trip**, every formula with its real numbers (Section 7), and where every formula comes from and how its numbers were set (Appendix B).

How to read this

Each section builds on the one before. Blue boxes explain an idea in plain words. Orange boxes are warnings. Green boxes are decisions made. Purple boxes are formulas, which you can skip. Every number here is read from the project's result files when this document is built, so it matches the poster and the abstract exactly.

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1. The problem in one page

Military vehicles often have to leave the road: to get around a blocked bridge, to reach a position with no road to it, or because the road is too exposed. Off the road, the ground decides whether a vehicle can move. Dry, firm soil carries a heavy truck easily. The same soil after three days of monsoon rain can turn to mud that swallows the wheels.

Whether a piece of ground can carry a particular vehicle is called **trafficability**. Planners describe it with three labels:

Label	Meaning
GO	The vehicle crosses normally, and many vehicles can follow in its tracks.
SLOW-GO	It gets across slowly, leaving ruts. Only a few vehicles can use the same track.
NO-GO	The vehicle gets stuck (or cannot climb, or cannot cross water).

Two things make this hard. **It changes over time:** the ground is firm in May, soft in July and firm again by November, so a map of where vehicles can go is right for one day only. **It depends on the vehicle:** a light tracked carrier crosses ground that would trap a wheeled armoured car.

> **The question this project answers**
>
> **A vehicle (or a convoy) has to cross the Pachmarhi plateau in the next few days. Which route avoids getting stuck, given that the ground will be wetter or drier by the time it gets there?**

Earlier work plans a route on the conditions *today*. We plan on a *forecast* of the travel day, made on the vehicle itself, and then check against two past monsoons whether that avoids routes that looked fine on planning day but failed later.

Pachmarhi is a hill station on the Satpura plateau in Madhya Pradesh, about 1,067 m above sea level. Most of its rain falls from June to September, so the ground swings from very firm to very soft every year. That makes it a good test site.

2. Ideas you need first

2.1 How firm the ground is: the cone index

Engineers measure ground firmness with a **cone penetrometer**: a rod with a small cone on the tip, pushed into the soil. The harder you have to push, the firmer the ground. The **rating cone index (RCI)** is a version of that reading that also allows for how much the soil weakens when vehicles knead it. Higher RCI means firmer ground. In this project RCI is the single number that says how firm the ground is on a given day.

2.2 What a vehicle needs: the vehicle cone index

Every vehicle has a **vehicle cone index (VCI)**, the firmness it needs to get through. Heavy vehicles on narrow tyres need firm ground; light vehicles on wide tracks manage softer ground. **VCI_1** is what one vehicle needs to pass once. **VCI_{50}** is what fifty vehicles need to use the same track, and it is higher, because each vehicle churns the soil for the next.

> **The rule that turns firmness into GO / SLOW-GO / NO-GO**
>
> - RCI at least VCI_{50}: **GO**
> - RCI between VCI_1 and VCI_{50}: **SLOW-GO**
> - RCI below VCI_1: **NO-GO** (the first vehicle gets stuck)

> **Decision: units are psi**
>
> The 10 October version warned that the code labelled cone index in kPa when the numbers were really in pounds per square inch (psi). The vehicle formulas (Section 6) produce psi, the field readings we rely on are in psi, and everything is now labelled psi.

2.3 Terrain

- **Slope:** how steep the ground is, in degrees.
- **Land cover:** forest, water, grass, crops, bare soil, buildings. A wheeled vehicle cannot drive through dense forest.
- **Wetness index (TWI):** a number computed from the elevation map that is high where water collects: low, flat places that drain the land around them [7].

2.4 Forecasting terms

- **Horizon:** how far ahead we predict; "h = 7" means seven days.
- **Persistence:** the lazy forecast "tomorrow is the same as today". Any real model must beat it.
- **RMSE:** the typical size of a forecast's mistake. **Skill:** how much smaller it is than persistence's.
- **Training / validation / test:** learn on 2015–21, tune on 2022–23, grade on 2024–25, which the model never saw.

2.5 Judging a decision: two kinds of mistake

A NO-GO warning can be wrong two ways. Missing a real NO-GO day sends vehicles out to get stuck (low **recall**). Warning on a day that was fine keeps a unit parked for nothing (low **precision**). You can always catch every NO-GO day by warning every day, so both must be reported together.

Routes have the same pair. A planner can send a vehicle along a route that turns out to be NO-GO (**stuck**), or find no route when one existed (**held back**). Every route result in this document reports both.

2.6 Route planning on a grid

The map is cut into 90 m square cells, each with a cost: cheap if the ground is easy, expensive if it is slow, forbidden if it is NO-GO. A route is a chain of neighbouring cells and its cost is the sum. **A*** and **Dijkstra's algorithm** both find the cheapest route exactly. We use Dijkstra, because one run gives routes from a start to every destination at once, which the backtest needs. A* would find the same routes.

2.7 Fuzzy sets

A fuzzy set lets things belong partly, with a degree between 0 and 1: a 12° slope can be "passable" to degree 0.55 [8]. Fuzzy AND is usually the **minimum**: "passable by slope AND by soil" is only as true as the weaker of the two. That is why a lake cell scores 0 however good its slope is.

2.8 Edge computing

Running software on a device in the field (a vehicle, a drone, a computer at a forward base) rather than in a distant data centre. It matters when the network is slow, unreliable or jammed, which is normal in military operations.

3. The conference: what ICDCN wants

ICDCN is the International Conference on Distributed Computing and Networking. The 2027 edition (the 28th) is at NITK Surathkal, 4–7 January 2027, and has a poster track for early-stage work.

Item	Detail
What you submit	A **2-page extended abstract**, two-column ACM format, fonts embedded, through EasyChair (track "ICDCN 2027 — Poster")
What you print	The poster itself, shown on an easel if accepted. ICDCN sets no size; we used **A0 portrait**, the usual default (MobiSys requires A0, COMSNETS accepts A0)
Review	Single-blind: reviewers see author names, so names go on page 1
Dates	Submit by **15 October 2026**. Notification 7 Nov. Camera-ready and registration 14 Nov
Judged on	"Technical merit and innovation as well as the potential to stimulate interesting discussions"
Presentation	In person only. No remote presentation
Publication	ACM Digital Library. ACM is now fully open access: authors at non-member institutions may owe a fee unless waived

3.1 The catch: this is a computing-and-networking conference

A poster that only says "we forecast soil strength with machine learning" sits outside ICDCN's main subjects. Applied work does get in when it ties to the call's topics, which include machine learning applications, edge computing, cyber-physical systems and vehicular networks.

> **Decision: how we frame the work**
>
> The planner runs **on a vehicle with no weather service and no network**. We back that with systems numbers: 531k parameters, 12 ms per forecast on one CPU core, and 18 bytes to send a 7-day forecast. The convoy experiment adds a genuinely distributed problem: vehicles planning on a shared map that they change as they drive, and exchanging only their tracks.

4. The data, file by file

4.1 The weather record

One row per day from 1 January 2015 to 31 December 2025 (4,018 days, none missing), with rainfall, temperature, humidity, wind, pressure, cloud cover and, most importantly, **soil moisture in two layers** (0–7 cm and 7–28 cm).

- Rainfall for 2015–2024 comes from the India Meteorological Department (IMD) gauge grid. 2025 comes from ERA5, because IMD's archive lags about a year.
- Soil moisture and temperature come from **ERA5-Land** [4], a computer model of the land surface steered by observations, on a grid cell about 9 km across.

> **Watch out**
>
> ERA5-Land soil moisture is a model output, not a measurement. Everything built on it inherits that.

4.2 From weather to ground firmness

The weather file has no trafficability column, so the project derives one. ERA5-Land has already turned rain and evaporation into soil moisture. We combine the two layers into a soil wetness index (SWI), turn that into RCI, and compare it with each vehicle's VCI.

> **The exact formula**
>
> $SWI = 0.35 \cdot s_{0–7\ \mathrm{cm}} + 0.65 \cdot s_{7–28\ \mathrm{cm}}$, each layer scaled to 0 (driest) – 1 (wettest) over the 11 years.
>
> $RCI = 300 \cdot e^{-2.996 \cdot SWI} \cdot (1 - 0.15 \cdot \min(R_{3\ \mathrm{days}} / 50\ \mathrm{mm}, 1))$ psi.
>
> The deeper layer weighs more because vehicles sink about 15–30 cm. The last factor briefly lowers firmness after rain, by at most 15%.

> **Decision: the wettest ground was lowered from 25 to 15 psi**
>
> With 25 psi as the wettest possible value, the ground never got softer than about 22 psi. Most light and medium tracked vehicles need less than that, so 695 of 1,004 of them could *never* be NO-GO. That was an artefact of one assumed number, not a finding. Roșca et al. [2] measured 14–25 psi on the field lanes where their 4×4 bogged down, so the wettest value is now 15 psi. Every vehicle band now has real NO-GO days (10–30% of days), and the forecaster was retrained on the new target.

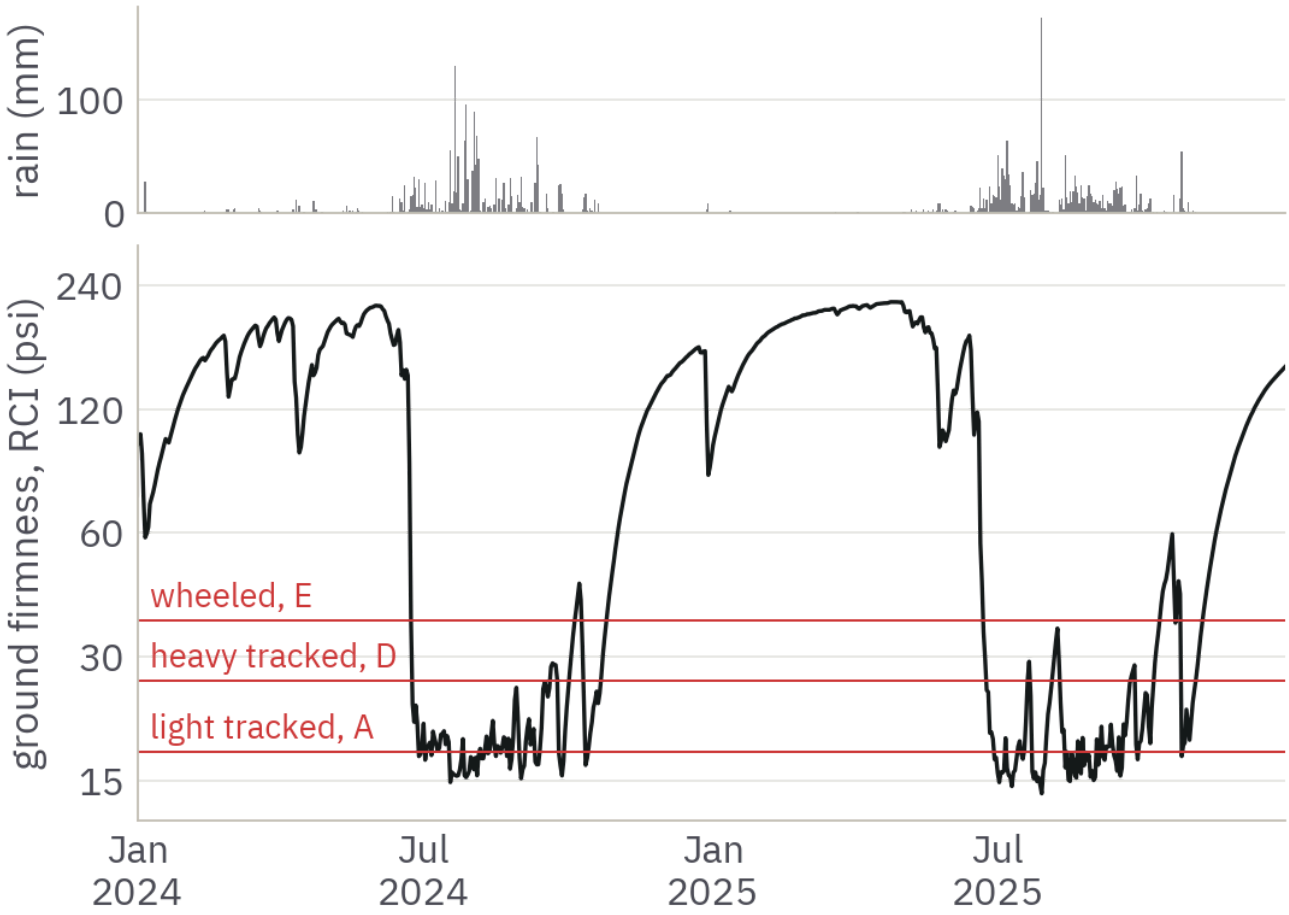


Figure 1. Rainfall (top) and ground firmness, RCI (bottom), over the 2024–25 test years. Each monsoon takes the ground into NO-GO within days. Red lines are the one-pass limits of three vehicle bands.

4.3 The satellite maps: your Earth Engine export

The GeoTIFF export arrived: three files, one per season (monsoon, summer, winter), each a 17-layer stack on one 30 m grid of 945 × 761 cells, about 28 × 23 km, in UTM zone 43N. It holds exactly what was asked for, and more:

Layers	Source	Used for
Elevation, slope, aspect, roughness (TRI)	Elevation model	Slope scores; wetness index
ESA WorldCover land cover [5]	Sentinel-1/2	Land-cover scores
NDVI, NDWI, bare-soil and snow indices	Sentinel-2	Canopy density (NDVI)
Radar VV, VH, ratio, 4 texture layers	Sentinel-1	Not used (see below)
Land surface temperature	Thermal	Not used

The parquet tables beside them hold the same pixels without map coordinates, so the project reads the GeoTIFFs. The old `maps.zip` pictures are kept but no longer used. The road network comes from OpenStreetMap [9]: 403 road segments a vehicle can use (footpaths and steps dropped).

Two things the data showed

- **About 79% of the area is tree cover.** Liu et al. score forest as impassable for every vehicle, which would close almost the whole plateau. Tree cover is therefore split by canopy density into Liu's forest, bush-wood and grassland rows (Section 6, Step 2).
- **Radar is not used.** Under this much canopy the radar signal mostly measures vegetation, not soil wetness, so using it to spread soil wetness would be a guess.

4.4 The vehicle file

2,451 rows, one per vehicle, with 34 columns such as mass, engine power, armour and weapons.

What this file actually is

A datamine of the video game War Thunder (game-engine fields such as `bone_gun`, joke entries such as "santa" and railway wagons). Two problems turned out to be bigger than first thought:

- **Masses are game templates.** Of 1,265 rows with a believable mass and power, there are only 90 distinct masses, and 68% of rows sit on 15 of them. The game copies one weight to many vehicles: the M1 Abrams' 54 t is right, a T-72's 50 t is not.
- **160 rows carry exactly 5,500 kg**, which is the game's default for wheeled vehicles. So wheeled armour (BTR-80, Boxer, Centauro and so on) has no real mass in the file.

Decisions: which vehicles, and how to treat the masses

- **Scope:** every usable ground vehicle, tracked, wheeled and half-track. The earlier reading ("Class B", then "wheeled only") was wrong.
- **Kept:** 1,145 vehicles: 1,004 tracked, 125 wheeled and 16 half-tracks. Wrecks, tutorials, event copies, scenery and joke entries were dropped.
- **Masses:** 111 placeholder rows were corrected from rounded public figures for about 60 base models, each marked in the table. Placeholder rows with no match were dropped.
- **Results are reported by firmness band, never by named vehicle**, because a name may carry a template mass. A band is a real range of vehicle needs, whichever names fall in it.

Corrections to the 10 October version

The ZIS-30, listed as a "truck", is a tracked tank destroyer on an artillery-tractor chassis. The ZIS-43 is a half-track. Classification is now done by name with checked keyword lists.

Band	VCI_1 range (psi)	Mostly	Vehicles	Typical VCI_1 / VCI_{50}	Days NO-GO, 2015–25
A	0–18	light tracked	209	17.6 / 41.5	10%
B	18–21	light-medium tracked	321	19.0 / 44.5	15%
C	21–24	medium tracked, some half-tracks	199	21.6 / 50.2	20%
D	24–30	heavy tracked, some wheeled	347	26.2 / 60.1	24%
E	30+	medium and heavy wheeled	69	36.7 / 86.2	30%

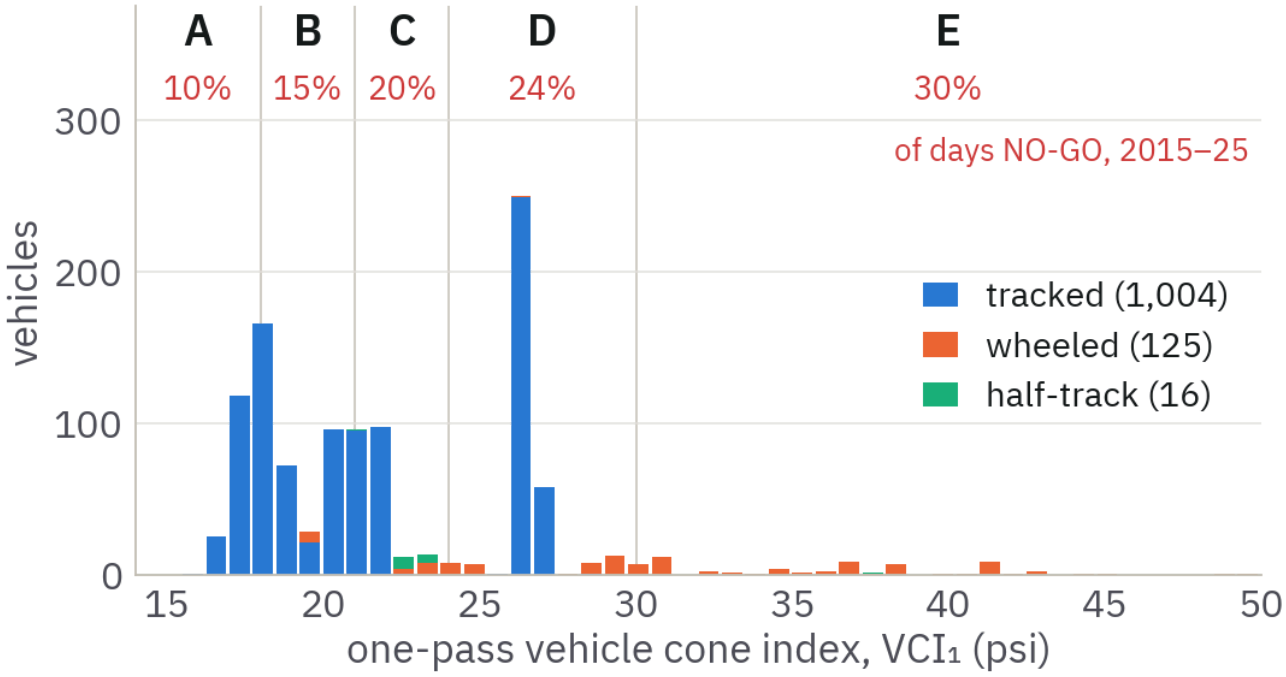


Figure 2. One-pass vehicle cone index of every vehicle, coloured by type, with the five bands and how often each band is stuck.

4.5 The inspiration paper: Liu et al. (2025)

Construction of dynamic trafficability map for unmanned vehicles considering multiple environmental factors and path planning, Scientific Reports 15:9957 [1]. They scored 11 factors (slope, land cover, rainfall and others) from lookup tables, combined them with a weighted average whose weights came from expert comparisons (AHP), forced any cell that is NO-GO in one factor to NO-GO, and planned routes with A* for one wheeled and one tracked vehicle.

What we take from them: their slope and land-cover tables, and their weights as the comparison baseline. **What we add:** planning on a forecast of the travel day instead of the conditions at the time, checking the routes against what really happened, a fleet of 1,145 vehicles instead of two, and vehicles that change the ground for each other.

5. The forecaster

5.1 The trick that makes learning work

Ground firmness changes slowly: today's value is almost exactly yesterday's. A model asked to predict tomorrow's *level* learns to copy today. So every model predicts the **change** from today, and today's value is added back afterwards.

5.2 The model, retrained on the new target

The September comparison chose a GRU (a recurrent neural network, 128 units in 2 layers) averaged with a ridge regression. Width helped more than architecture; attention and bidirectional variants made it worse. That model was retrained for the new firmness target (Section 4.2), with three GRUs from different random starts averaged.

Model	RMSE, 1 day (psi)	3 days	7 days	Skill at 7 days
GRU + ridge, averaged (chosen)	**6.05**	**13.29**	**20.76**	**+20%**
GRU alone	6.14	13.98	21.85	+16%
Ridge alone	6.31	13.86	21.92	+16%
Persistence ('same as today')	6.71	15.16	26.02	+0%

> **The September numbers no longer apply**
>
> The model comparison in `outputs/forecasting` (the ensemble table, the TFT results) was computed with the old firmness target. Its conclusions about which model to use stand; its numbers must not be quoted. The figures above are the current ones.

5.3 From a forecast to a NO-GO call: safety margins

The model's average error rewards predicting values near the middle, and NO-GO days are the extreme. So the forecast is lowered by a **safety margin** before it is compared with VCI_1, and borderline days get called NO-GO. There is one margin per firmness band and lead time, set to the smallest that catches 90% of NO-GO days.

> **Decision: margins are tuned on one year and scored on the other**
>
> A margin tuned and tested on the same days always looks good. Each margin is tuned on 2024 and scored on 2025, then the other way round, and the two results are pooled.

Band (7 days ahead)	Today's value + margin: recall / false alarms per vehicle-year	Forecast + margin: same	Margin tuned on 2024 / 2025 (psi)
A	89% / 74	91% / 54	13.8 / 7.0
B	88% / 54	91% / 34	10.8 / 9.2
C	90% / 44	90% / 28	10.8 / 17.8
D	91% / 29	90% / 22	19.5 / 14.8
E	90% / 14	90% / 8	6.0 / 9.8

At the same catch rate, the forecast raises 23–45% fewer false alarms at 7 days than putting the same kind of margin on today's value.

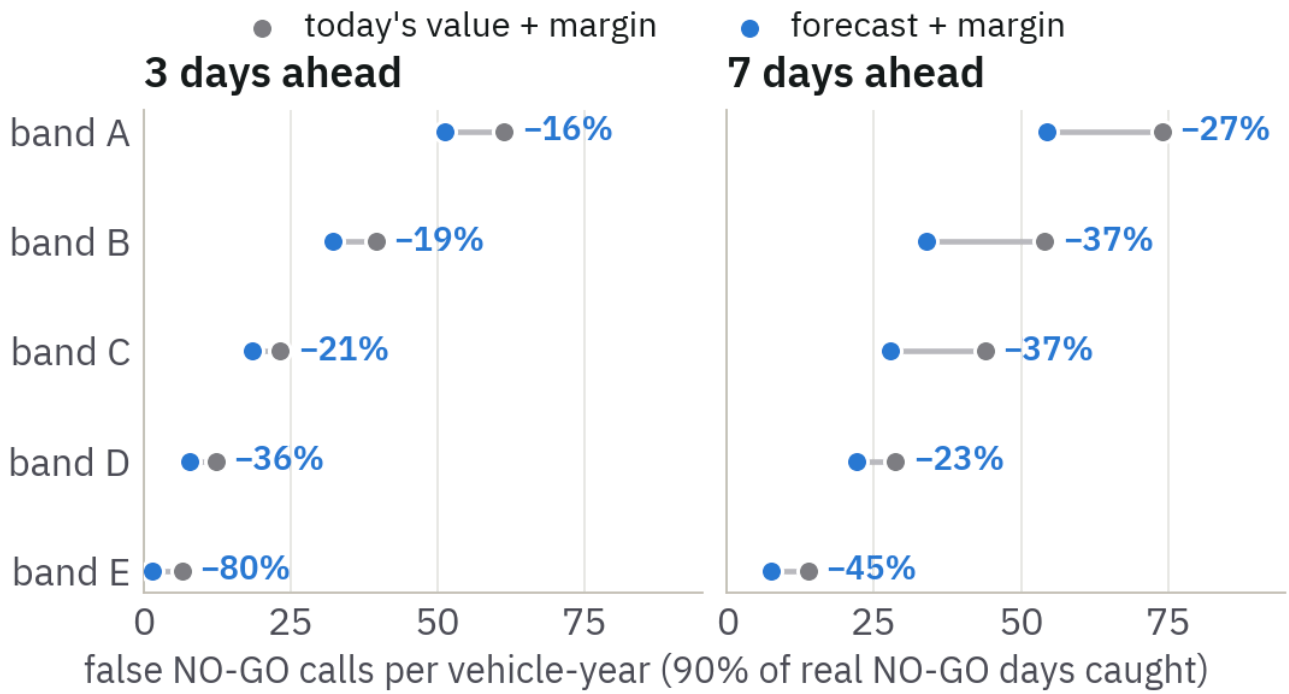


Figure 3. False NO-GO calls per vehicle-year, 3 and 7 days ahead, with about 90% of real NO-GO days caught in every case.

Margins move between years

The same band's margin can differ by up to 7.2 psi depending on which year it was tuned on. The cross-year scoring above already includes that cost, but a field system would need margins re-checked as more years come in.

Still true from the 10 October version: do not quote the TFT's 91.4%

The TFT model's "P10" rule reached 91.4% NO-GO recall by calling NO-GO far too often, and its margin was unstable across years. It was rejected in September, and the README now says so.

5.4 What the forecast cannot see

The forecaster sees only the local weather log, not a weather forecast. Nothing in that log announces the day the monsoon breaks, so on the first days of the monsoon every planner gets caught out. The gain comes in the days around the transitions, when the ground keeps changing and the forecast tracks it.

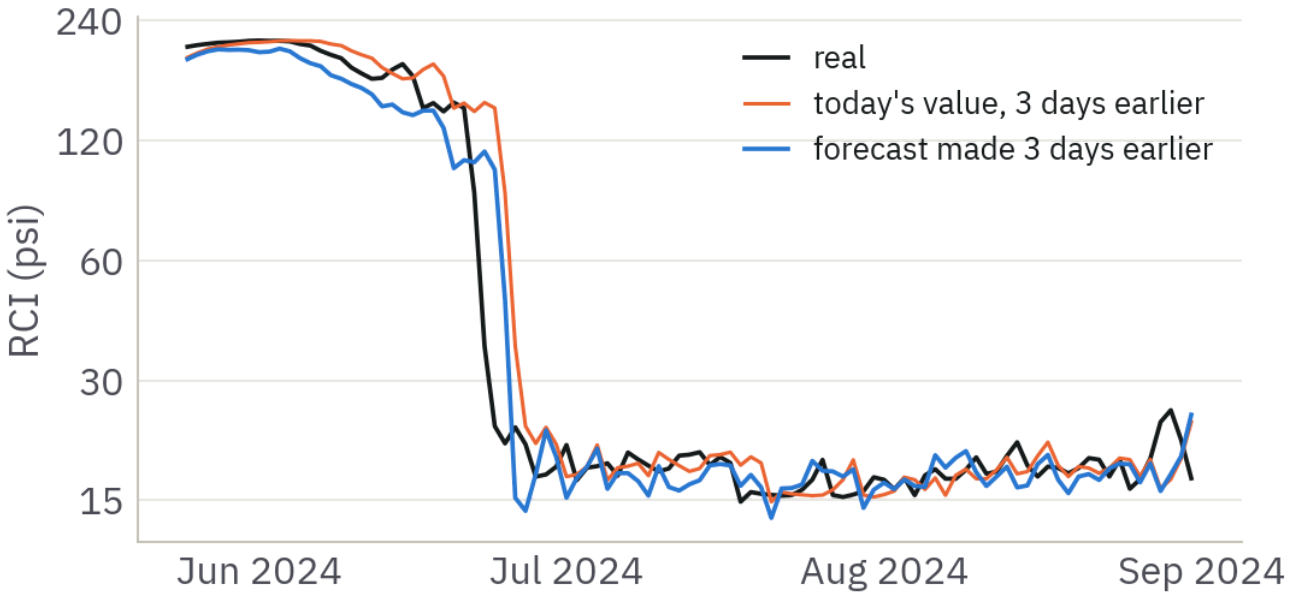


Figure 4. Monsoon onset, 2024: real firmness (black) against today's value and the forecast, both made three days earlier.

6. What was built, step by step

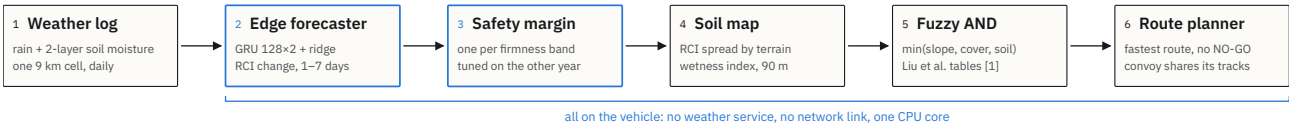


Figure 5. The pipeline. Steps 2–6 run on the vehicle.

Step 1. Vehicle profiles

Each vehicle needs VCI_1 and VCI_{50}. They come from the US Army's **Mobility Index** method (NRMM; test procedure TOP 02-2-619A), which turns weight, track or tyre size, ground clearance, power and gearbox into one number. The Mobility Index tables and all four MI-to-VCI curves come from Rula and Nuttall [12] and the NRMM Edition I users guide [13]; the wheeled additions of NRMM II (weight factor above 31,500 lb, VCI_1 above MI 115, tyre deflection) follow Roșca and Ciobotaru [14] and Roșca et al. [2], and the tracked worked example is Ciobotaru's [3].

The formulas

Tracked: MI = (CPF·WF / (TEF·GF) + BF − CF) · EF · TF, with CPF = W / (2·L·B), TEF = B/100, BF = W / (10 · road wheels · shoe area), CF = clearance/10. VCI_1 = 7.0 + 0.2·MI − 39.2/(MI + 5.6); VCI_{50} = 19.27 + 0.43·MI − 125.79/(MI + 7.08).

Wheeled: MI = (CPF·WF / (TEF·GF) + WLF − CF) · EF · TF, with CPF = axle load / (0.5 · tyres per axle · diameter · width), TEF = (10 + width)/100, WLF = axle load/2000. VCI_1 = 11.48 + 0.2·MI − 39.2/(MI + 3.74) up to MI 115, 4.1·MI^0.446 above; VCI_{50} = 28.23 + 0.43·MI − 92.67/(MI + 3.67). The engine factor is 1.05 below 10 hp per short ton (2,000 lb). Loads in lb, lengths in inches, VCI in psi.

Checked against published examples

With the M1 Abrams' public specifications, the tracked code gives MI 98.61, VCI_1 26.35 and VCI_{50} 60.48. Ciobotaru [3] reports 98.59, 26.34 and 60.48. For the six 8×8 personnel carriers of Roșca and Ciobotaru [14], the wheeled code matches 22 of the 24 published VCI values to the nearest psi; the other two (one vehicle's VCI_{50}) are within 1.7 psi.

What had to be assumed. The file has no track or tyre sizes, so they come from weight: tracks interpolated between real vehicles from the M22 Locust to the Tiger II, tyres by the load each axle carries. Transmissions are taken as automatic and tyres at road pressure. Half-tracks have no standard formula: the track unit is rated as tracked, the front axle as a wheeled axle carrying 25% of the weight, and the worse of the two governs.

Step 2. Terrain scores

Each 90 m cell gets a 0–1 score for slope and for land cover, per vehicle type, from Liu et al.'s tables. The slope steps are smoothed into a curve, so a 0.2° change cannot flip a decision. Tree cover is split

by canopy density, using each cell's greenest NDVI across the three seasons. The plateau forest is deciduous, so one season's NDVI tracks leaves, not trees.

- Dense canopy (NDVI ≥ 0.75): Liu's "forest", 0 for every vehicle.
- Open canopy (0.60–0.75): "bush-wood", 0 wheeled, 0.2 tracked.
- Sparse (below 0.60): "grassland", 0.4 wheeled, 0.6 tracked.
- Water is 0 for all. Road cells score 1 for slope and land cover.

Before any soil is considered, 26% of the plateau is passable for wheeled vehicles (mostly roads and open land) and 72% for tracked ones.

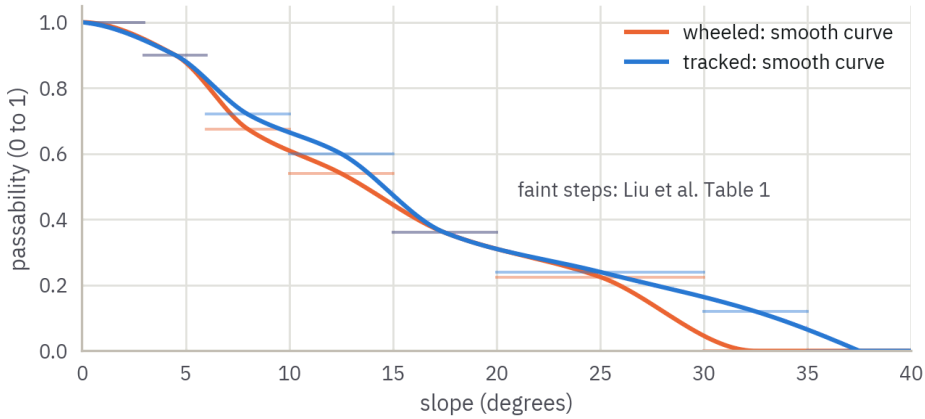


Figure 6. Liu et al.'s slope steps (faint) and the smooth curves the planner uses.

Step 3. Spreading firmness across the map

The forecast is one value for a 9 km weather cell, which on its own would make the whole map wet or dry at once. Each cell's soil wetness is shifted by its wetness index: low, convergent cells up to 0.2 wetter, ridges up to 0.2 drier, the average unchanged. In the firmness formula that is $RCI_{cell} = RCI \cdot e^{-2.996 \cdot shift}$. Paved roads ignore soil; unpaved tracks soften like any ground.

> **A rule of thumb**
>
> This spreading has no field measurements behind it. The poster and abstract say so.

Step 4. Combining into one score

Soil scores 0 below VCI_1 (NO-GO) and rises to 1 at VCI_{50}, so SLOW-GO ground is slow but allowed. The three scores combine by **fuzzy AND, the minimum**. A cell is as passable as its worst factor, so a deal-breaker can never be averaged away. Step 8 turns the SLOW-GO range into two plain numbers: vehicles per track and km/h.

Step 5. Planning routes

The planner finds the fastest route, with speed proportional to the cell's score, on a grid of 253 × 315 cells. Five planners are compared:

Planner	Map it uses
Terrain only	Slope and land cover, no soil
Today's map	Today's ground, which is what Liu et al. do
Today + margin	Today's ground minus a safety margin
Forecast + margin (ours)	The forecast for the travel day minus the band's margin
Oracle	The real ground on the travel day, used only to score the others

"Speed proportional to the score" makes the route cost a relative time, not minutes. Step 8 adds a speed model in km/h, and the same maps can be routed by real travel time instead.

Step 6. The backtest

For every day of 2024–25 (724 planning days), each of the five bands, and departures 1, 3 or 7 days after planning: 30 trips of 8–25 km between points on the plateau's road network. That is 325,800 trips per planner. Every route is checked against the real ground on the travel day.

Planner: stuck / held back	1 day	3 days	7 days
Terrain only (no soil)	35.5% / 0.0%	35.5% / 0.0%	35.5% / 0.0%
Today's map (what Liu et al. do)	0.8% / 1.1%	2.1% / 3.0%	4.0% / 5.8%
Today's map + safety margin	0.7% / 1.5%	1.5% / 4.4%	2.5% / 8.4%
Forecast + safety margin (ours)	**0.7% / 0.7%**	**1.6% / 2.3%**	**2.9% / 4.5%**

What the backtest supports, and what it does not

- **Against today's map:** at 3 days, 1.6% of forecast-planned trips got stuck against 2.1% (24% fewer), and fewer were held back (2.3% against 3.0%). At 7 days, 27% fewer stuck.
- **Against a margin on today's map:** the forecast does *not* get fewer trips stuck at the tuned setting. What it does is hold back far fewer trips at the same risk. Turning each planner's margin from none up to three times the tuned value (Figure 7), the forecast holds back 40% (3 days) and 31% (7 days) fewer trips at an equal stuck rate.
- **Cost:** forecast-planned routes are only 0.06% slower than the best possible route.
- Ignoring soil altogether sends 36% of trips into mud.

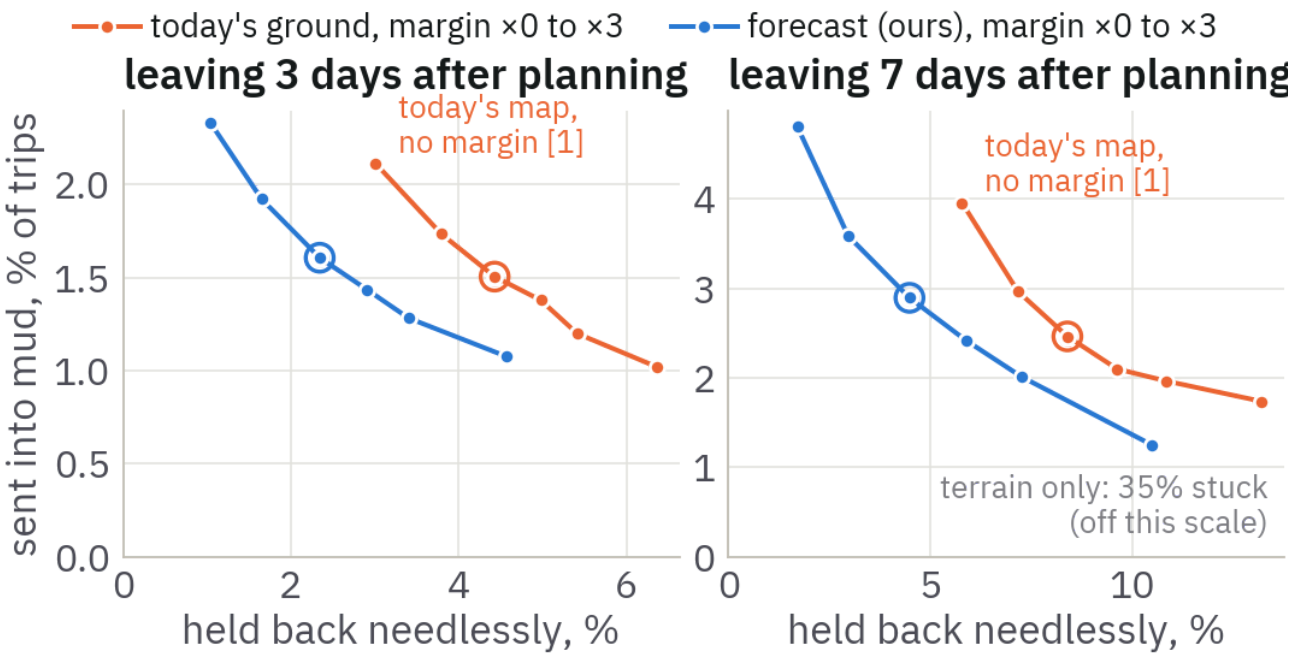


Figure 7. Both errors as each planner's margin goes from none to three times the tuned value (ringed). Lower left is better.

Combining slope, cover and soil: stuck	1 day	3 days	7 days
Fuzzy AND, minimum (ours)	**0.7%**	**1.6%**	**2.9%**
Fuzzy AND, product	0.7%	1.6%	2.9%
Weighted mean + veto (Liu et al.)	0.9%	1.8%	3.0%
Weighted mean, no veto	34.6%	34.8%	34.9%

Liu et al.'s weighted mean needs their veto to work: without it, a third of trips go into mud. With the veto it is close to the minimum, slightly worse.

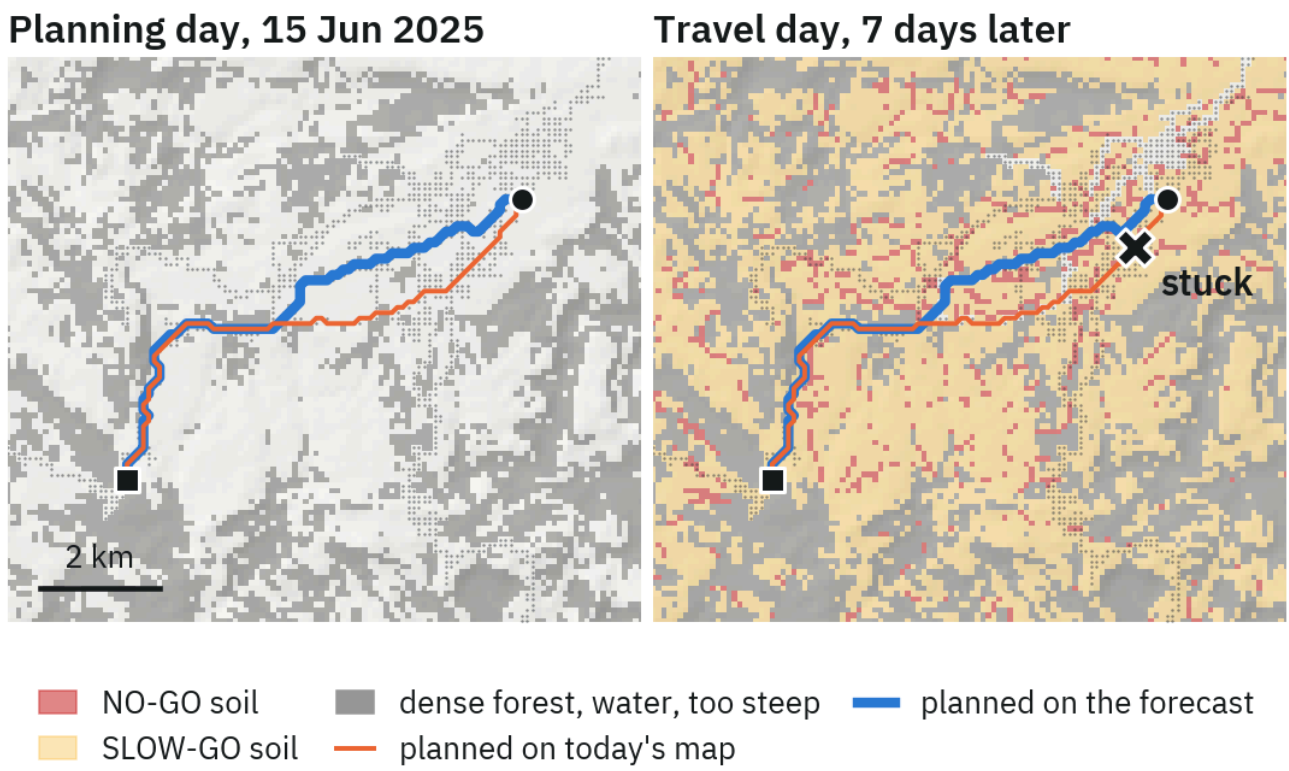


Figure 8. One real trip from the backtest. A tracked vehicle from band D (VCI_1 26 psi) plans on 15 June 2025 for a trip 7 days later. On the planning day the ground reads 128 psi and the direct route (orange) looks open. On the travel day it reads 34 psi and that route crosses NO-GO soil (×). The route planned on the forecast (blue, margin 19.5 psi) avoids it. Circle: start; square: end; grey lines: OpenStreetMap roads.

Step 7. Vehicles changing the ground: convoys

Every vehicle that crosses soft ground churns it, so the next one needs firmer ground. Each cell counts how many vehicles have crossed it.

> **The wear rule**
>
> Firmness needed on pass n = VCI_1 + (VCI_{50} − VCI_1) · ln(n) / ln(50). The curve is our own assumption, chosen so that pass 1 gives VCI_1 and pass 50 gives VCI_{50}; it is not a published law. The start and finish areas are not churned, because every vehicle has to use them.

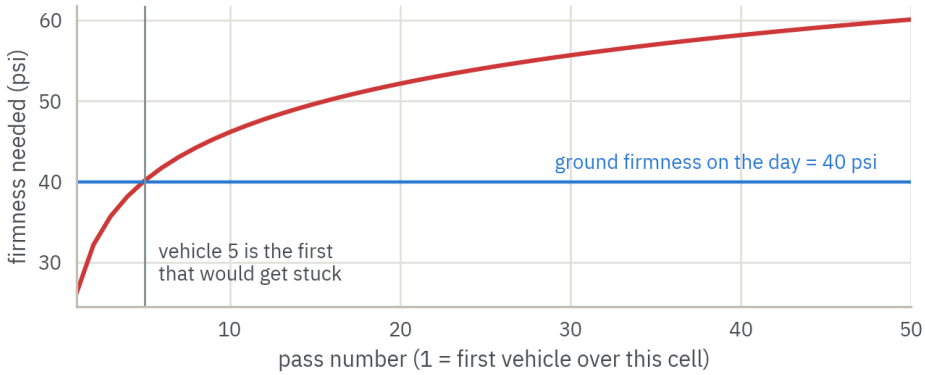


Figure 9. Heavy tracked band D (VCI_1 26.2, VCI_{50} 60.1 psi): on ground of 40 psi, vehicle 5 would be the first to get stuck on a shared track.

Two ways of moving ten vehicles, on 54 soft days:

- **Follow the leader:** everyone takes the first vehicle's route; when one gets stuck the column behind it is blocked.
- **Wear-aware:** each vehicle plans on the shared pass-count map and broadcasts its own track (4 bytes per cell).

Heavy tracked vehicles: **5.5** of 10 get through following the leader, **9.2** with the shared map, for 7.2 kB of track updates per convoy. Wheeled vehicles: 6.8 and 7.3. In this forest they have few parallel lanes to spread onto.

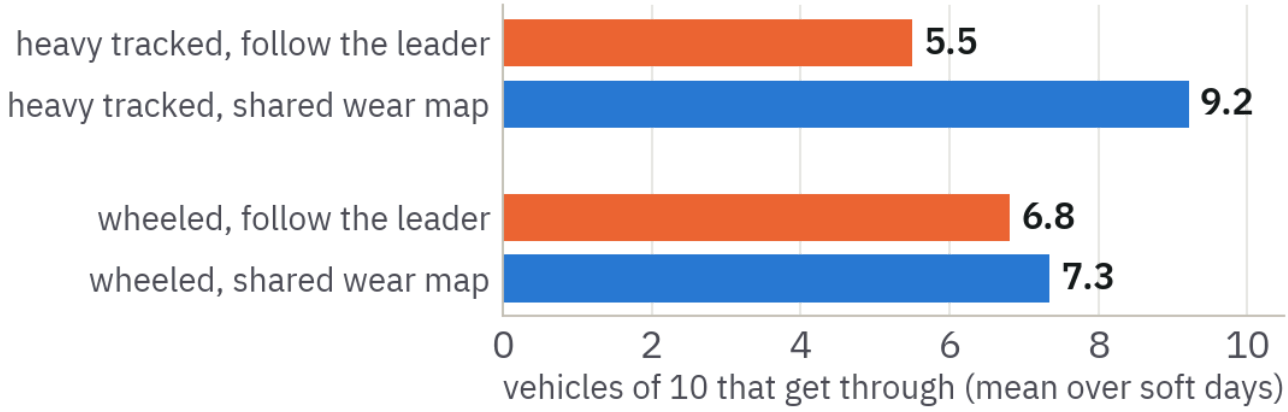


Figure 10. Vehicles of ten that get through, by band and convoy method.

Step 8. SLOW-GO in numbers: passes and speed

SLOW-GO covers everything from "one vehicle gets across, barely" to "forty-nine vehicles can use the same track at a decent pace". A commander needs to know where in that range the day falls. Two numbers now say it: how many vehicles one track carries, and how fast the band can drive.

Passes: how many vehicles one track carries

> **The idea**
>
> VCI_1 is the firmness one vehicle needs; VCI_{50} is the firmness fifty vehicles need on the same track. Every pass churns the soil, so each vehicle needs a little more firmness than the one before it. Ground that sits partway between VCI_1 and VCI_{50} carries a number of vehicles partway between 1 and 50. This is the convoy wear rule of Step 7 turned round: Step 7 asks "how firm must the ground be for vehicle number n?", this asks "how many vehicles can this ground take?".

> **The formula**
>
> passes $n = 50^{(RCI - VCI_1) / (VCI_{50} - VCI_1)}$
>
> It gives 1 at VCI_1, 50 at VCI_{50} (GO), and 0 below VCI_1 (NO-GO). It is the exact inverse of the wear rule, firmness needed = $VCI_1 + (VCI_{50} - VCI_1) \cdot \ln(n)/\ln(50)$.

Our assumption

The published sources give only the two end points, 1 pass and 50 passes [12]. The logarithmic curve between them is ours, shared with the wear rule.

Speed: how fast the band can drive

The speed comes from the soil relations of the NATO Reference Mobility Model (NRMM), the model the US Army and NATO use to predict cross-country speed [13]. Its fine-grained soil relations were fitted to more than 1,600 field tests with wheeled and tracked vehicles [12]. In plain words, a vehicle on soft ground faces three things:

- **Excess firmness**, x = RCI − VCI_1: how far the ground is above the vehicle's limit. Everything else is a function of it.
- **Motion resistance**, R/W: the drag of sinking and pushing soil, as a fraction of the vehicle's weight. It is about 0.40 at VCI_1 (pulling 40% of your own weight just to keep moving) and falls to about 0.05 on firm ground.
- **Drawbar pull**, D/W: the pull left over after that drag. It is 0 at VCI_1, which is what VCI_1 means: the vehicle can just move itself and nothing more.

On a slope the vehicle also has to lift its own weight, so the force it needs is the soil drag plus the slope. If the soil cannot grip hard enough to supply that force, the vehicle is stuck on that slope even though the soil is above VCI_1. Otherwise the engine sets the speed: power is force times speed, so speed = engine power ÷ force needed. Finally the tracks or wheels slip a little in soft soil, which takes some of that speed away.

The formulas (NRMM Edition I, fine-grained soil [13])

x = RCI − VCI_1

Drawbar pull at 20% slip: D/W = $a + bx - \sqrt{((a + bx)^2 - cx)}$ for $x \geq 0$; $s \cdot x$ below.

Motion resistance: R/W = $p + q / (x + r)$ for $x \geq 0$; $u - s'x$ below.

Most traction the soil can give (at 100% slip): CF = D/W − k_1 + R/W, TFOR/W = CF + k_2.

Force needed on a slope of angle θ: load = sin θ + R/W · cos θ. Stuck if TFOR/W · cos θ < load.

Slip: $S = \alpha y + \beta + \gamma/(\delta - y)$, with y = load / cos θ − CF.

Speed: $v = \eta \cdot P / (W \cdot \text{load}) \cdot (1 - S)$, capped at the top speed. P engine power, W weight, η = 80% of the engine's power reaching the tracks or wheels.

Paved road ignores the soil: R/W = 0.045 for tracks, 0.007 + 0.0939/tyre pressure for tyres [13].

Running gear (contact pressure)	a, b, c, s	p, q, r, u, s′	k_1, k_2	α, β, γ, δ
Tracked, below 4 psi	.544, .0463, .0702, .076	.045, 2.3075, 6.5, .4, .072	.758, .82	.0257, −.0161, .01519, .8353
Tracked, 4 psi or more	.455, .0392, .0526, .056	.045, 2.3075, 6.5, .4, .052	.671, .71	.0733, −.0063, .00734, .7177
Wheeled, below 4 psi	.3885, .0265, .0358, .046	.035, .861, 3.249, .3, .043	.674, .76	.0621, −.021, .01888, .7794
Wheeled, 4 psi or more	.379, .0219, .0257, .033	.045, 2.3075, 6.5, .4, .029	.585, .655	.084, −.016, .01414, .6697

Two checks

- **It agrees with our NO-GO line.** At exactly VCI_1 the relations give 20% slip, the slip at which NRMM's VCI_1 is defined, and a level-ground speed above zero. On a slope at VCI_1 the vehicle is stuck, because there is no spare pull to climb with.
- **M1 Abrams:** the model gives 12 km/h at the M1's VCI_1 and 45 km/h at 20 psi above it. Ciobotaru's published NRMM run for the M1 [3] shows speeds of about 20 km/h just before NO-GO and 56 km/h at most, the same range.

Band	Type	hp/t	Contact pressure (psi)	km/h at VCI_1	+5 psi	+10	+20	+40	Vehicles per track at +5 / +10 / +20 psi
A	tracked	19.0	9.2	8	16	22	31	43	2 / 5 / 26
B	tracked	15.6	9.5	7	13	18	25	36	2 / 5 / 21
C	tracked	14.0	12.0	6	12	16	23	32	2 / 4 / 15
D	tracked	15.7	12.8	7	13	18	26	36	2 / 3 / 10
E	wheeled	19.2	20.0	8	16	22	31	44	1 / 2 / 5

Each band's planning vehicle takes the median power-to-weight and contact pressure of the band's vehicles of its type.

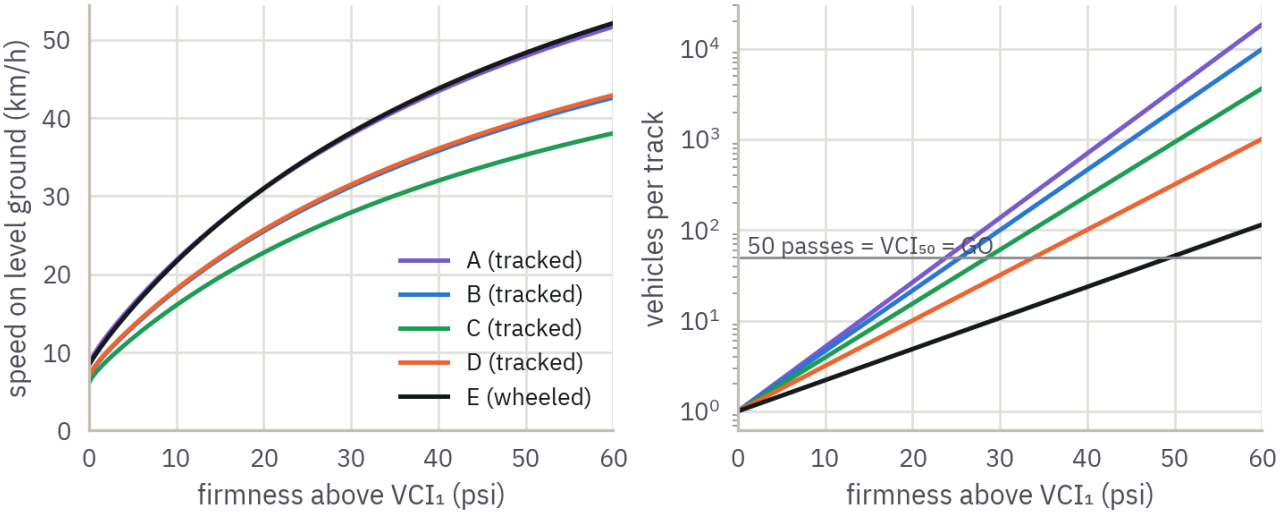


Figure 11. Left: level-ground speed of each band's planning vehicle as the ground firms up past its VCI₁. Right: vehicles one track carries; 50 is GO.

Routing by travel time

Every map cell now has a speed: the soil speed above, using the slope along the direction of travel (from the elevation of the two cells), so uphill is slower than downhill. Off the road the speed is scaled by the land-cover score, so open forest is slower than grass. Paved roads use road rolling resistance and ignore the soil. A step between two cells takes half its length at each cell's speed, and Dijkstra finds the route with the least total time. The result is a trip time in minutes.

Planner: stuck / held back / minutes per trip	1 day	3 days	7 days
Published planner (passability cost)	0.7% / 0.7% / 32.5 min	1.7% / 2.3% / 32.4 min	2.9% / 4.5% / 32.2 min
Routed by travel time	**0.8% / 0.7% / 31.0 min**	**1.6% / 2.4% / 30.8 min**	**2.9% / 4.5% / 30.7 min**

What the speed backtest shows

- **Same safety:** choosing routes by travel time changes stuck and held-back trips by at most 0.05 percentage points. Here "stuck" also counts climbs the soil cannot carry the vehicle up.
- **Faster trips:** routes are 5% quicker at 3 days.
- **Trip times can be planned:** the time predicted on the forecast map is +0.3% off the time on the real travel day at 3 days on average (+1.0% at 7), and 9 trips in 10 are within -2.1% to +3.4%. Times run slightly long, because the safety margin makes the planned ground a little softer than the real ground.

What had to be assumed for speed

- The vehicle file has engine power but no gearbox, so the engine is taken as giving the same power at every speed. NRMM builds the pull at each speed gear by gear.
- η = 80% of the engine's power reaches the ground. Speed scales with it: at 70% every speed is -12% lower.
- Top speed is the game's template value (60 or 75 km/h), not a real figure.
- NRMM also limits speed for tree stems, ride comfort and obstacles. None of these is modelled, so speeds are upper bounds; the land-cover score stands in for vegetation.
- The travel-day times come from the same model, so the backtest shows what forecast error does to trip times, not that the model's km/h are right in the field.

Step 9. Edge numbers

What	Value
Model size (3 GRUs of 176,775 parameters, plus ridge)	531k parameters, 2.0 MB
One 7-day forecast, one CPU thread	12 ms (+ 21 ms to build inputs)
Re-planning one route, 76k-cell map	19 ms
Re-planning one route by travel time	82 ms
Sending a 7-day forecast	18 bytes
Sending one passability map instead	76 kB (4,200× more)
Maps for 5 bands × 7 days	2.7 MB

Because every vehicle rebuilds its own maps from the forecast and the shared tracks, the radio link only ever carries the small things. Passes and speed are worked out on the vehicle from the same 18-byte forecast, so they add nothing to the link. In the edge bundle, `outlook()` gives status, passes and level-ground km/h for each of the next 7 days, and `plan(..., by="time")` routes by travel time.

7. One trip, end to end: every formula with real numbers

This follows one real planning day through every stage, from the weather log to the route and the check on the travel day. On **24 October 2024** the plateau was drying out after the monsoon: firmness was 33.2 psi. A band-D unit (heavy tracked) plans a trip for **27 October 2024**, three days later. Every number below is computed by the project's own code when this document is built; where a stage can be checked against another part of the pipeline, the build checks it.

Stage	Formula	With the numbers	Result
1. Soil moisture → wetness	$s = (\theta - 20) / (49 - 20)$, clipped to 0–1 $SWI = 0.35 \cdot s_{0-7} + 0.65 \cdot s_{7-28}$	θ = 40.3 and 41.9 vol% → s = 0.700, 0.754 $SWI = 0.35 \cdot 0.700 + 0.65 \cdot 0.754$	SWI = 0.735
2. Wetness → firmness today	$RCI = 300 \cdot e^{-2.996 \cdot SWI} \cdot (1 - 0.15 \cdot \min(R_{3d}/50, 1))$	R_{3d} = 0.0 mm → rain factor 1.000 $300 \cdot e^{-2.996 \cdot 0.735} \cdot 1.000$	RCI_{today} = 33.2 psi
3. Forecast for the travel day	$RCI_{t+3} = RCI_t + ½ \cdot (\Delta GRU + \Delta ridge)$	$33.2 + ½ \cdot (+6.99 + 12.40)$ 3 GRUs averaged: ΔGRU; 1 ridge regression: Δridge	$RCI_{forecast}$ = 42.9 psi
4. Safety margin	RCI_{safe} = forecast − m(band, lead)	m(D, 3 days) = 1.75 psi, tuned on 2025 42.9 − 1.75	RCI_{safe} = 41.1 psi
5. A band-D vehicle → MI	$MI = (CPF \cdot WF / (TEF \cdot GF) + BF - CF) \cdot EF \cdot TF$	50.0 t = 110,231 lb, 810 hp; tracks 24.4 × 176 in, 14 road wheels, clearance 18.8 in CPF = 12.82, WF = 1.8, TEF = 0.244, GF = 1, BF = 4.97, CF = 1.88, EF = 1.00, TF = 1	MI = 97.72
6. MI → vehicle cone index	$VCI_1 = 7.0 + 0.2 \cdot MI - 39.2/(MI + 5.6)$ $VCI_{50} = 19.27 + 0.43 \cdot MI - 125.79/(MI + 7.08)$	MI = 97.72	VCI_1 = 26.2, VCI_{50} = 60.1 psi
7. Band planning vehicle	median VCI_1, VCI_{50}, power-to-weight and contact pressure of band D's tracked vehicles	347 vehicles in band D	VCI_1 26.2, VCI_{50} 60.1 psi; 15.7 hp/t; CPF 12.8 psi
8. Decision	NO-GO if $RCI_{safe} < VCI_1$; SLOW-GO if $< VCI_{50}$; else GO	26.2 ≤ 41.1 < 60.1	**SLOW-GO**
9. Passes per track	$n = 50^{(RCI - VCI_1)/(VCI_{50} - VCI_1)}$	$50^{(41.1 - 26.2)/(60.1 - 26.2)}$	5.6 → **5 vehicles** per track
10. Soil traction (NRMM)	$x = RCI - VCI_1$ $D/W = a + bx - \sqrt{((a + bx)^2 - cx)}$ $R/W = p + q/(x + r)$	x = 15.0 psi; tracked, CPF ≥ 4 psi D/W = 0.496, R/W = 0.153 TFOR/W = D/W − 0.671 + R/W + 0.71 = 0.687	moves if TFOR/W ≥ R/W (0.687 ≥ 0.153)
11. Speed on level ground	$v = \eta \cdot P / (W \cdot R/W) \cdot (1 - S)$, capped at top speed	η = 0.8, P/W = 15.7 hp/t → power-limited 22.6 km/h slip S at y = R/W − CF = 0.175: 2.0%	**22.1 km/h**

12. Firmness in one map cell	shift = $0.10 \cdot \text{clip}(z_{TWI}, -2, 2)$; $RCI_{cell} = RCI \cdot e^{-2.996 \cdot shift}$	the route's weakest unpaved-track cell: z = +0.62 → shift +0.062	RCI_{cell} = 34.1 psi
13. Cell passability	min(slope score, cover score, soil score) soil = $0.25 + 0.75 \cdot (RCI - VCI_1)/(VCI_{50} - VCI_1)$	unpaved track: slope and cover count as 1 on roads; soil $0.25 + 0.75 \cdot (34.1 - 26.2)/(60.1 - 26.2) = 0.43$	μ = 0.43
14. Route (published planner)	Dijkstra on cost = $\text{step} \cdot \frac{1}{2}(1/\mu_u + 1/\mu_v)$	trip of 26.9 km on the grid, 253 cells	cost 29.52 (km ÷ speed fraction)
15. Speed in that cell	same as 10–11 with load = $\sin\theta + R/W \cdot \cos\theta$, × cover score	step into the cell: grade +8.2%, cell RCI 34.1 psi	11.5 km/h
16. Travel time	$\Sigma\ \text{step} \cdot \frac{1}{2}(1/v_u + 1/v_v)$	this route; the route chosen by travel time instead	46 min; 44 min
17. Check on the travel day	real RCI on the travel day, same map rules	27 Oct 2024: RCI 42.2 psi	route passable; 45 min (best possible 44)
18. What is radioed	4-byte day number + 7 × 16-bit firmness values	4 + 7 × 2	18 bytes

Reading the chain

- **Stages 1–4 run on the vehicle once a day** and serve every band: one firmness forecast, minus each band's margin.
- **Stages 5–7 are done once, in advance**: every vehicle's VCI, and one planning vehicle per band.
- **Stages 8–11 turn the forecast into the band's call**: SLOW-GO, how many vehicles a track carries, and how fast on level ground.
- **Stages 12–16 build the map and the route**: each cell's firmness, its passability, the route, and the trip time.
- **Stage 17 is the backtest's check**: the travel day's real firmness was 42.2 psi, and the route is scored against it. Stage 18 is what crosses the radio link.

8. Your questions, answered

8.1 Should we use JEPA?

Not for this poster. JEPA teaches an image model without labels by predicting hidden parts of images. It needs a large amount of imagery and a labelled task to show it helped. We have three seasonal snapshots of one area and no labels. It is a good future-work line: a JEPA-style model trained on years of Sentinel-1 and Sentinel-2 images could learn how the terrain changes over time.

8.2 Did the forecasting models need retraining?

Yes, but not because of the new data. The vehicle file and the maps still add nothing the forecaster can use, since it predicts one plateau-wide firmness from the weather log. It was retrained because the firmness target itself changed (Section 4.2). That took under a minute on the GPU.

8.3 Should we train a separate GO / SLOW-GO / NO-GO classifier?

No. The labels are firmness with cut-offs applied, so a classifier would learn a blurrier copy of the forecast. One forecaster serves every vehicle, because only the cut-offs change. There are no ground-truth labels for the map. The per-band margin does the decision job.

8.4 Can we model how vehicles change the terrain?

Yes, and it is built (Step 7): pass-count wear on a shared map. Ruts that persist and recover over days, and full soil physics, remain future work. Both need measurements we do not have.

8.5 Can we use fuzzy sets?

Yes, and it is built. Smooth curves replace Liu's step tables, and the fuzzy AND (minimum) combines factors without needing a veto. The comparison against Liu's weighted average is in Step 6.

8.6 Do the vehicles have to be Indian?

No. This is preliminary work. The poster uses every usable vehicle in the file, of any origin, reported by firmness band.

9. Limitations to be honest about

Limitation	Why it matters	What we do
Ground firmness is computed, not measured	No cone-penetrometer readings from Pachmarhi	State it; field readings are the first next step
Wettest firmness value (15 psi) from one field study	It sets how often light vehicles are stuck	Cite the study; note it on the poster
One 9 km weather cell	Real firmness varies over metres	Spread by wetness index; say it is a rule of thumb
Vehicle data from a video game	Masses are template values	Correct the worst; disclose; report bands only
Track and tyre sizes assumed by weight	They drive the cone index	Interpolate between real vehicles; state it
Margins move between years	By up to 7.2 psi	Score across years; state it
No weather forecast input	Monsoon onset cannot be foreseen	Show it openly (Figure 4)
Convoy wear curve is our assumption	Not a published law	Say so
Passes between 1 and 50 follow the same assumed curve	Only the two end points are published	Say so
Speed: same power at every speed, 80% efficiency, game top speed	No gearbox data; speeds scale with the efficiency	State it; report speeds as upper bounds
No vegetation, ride or obstacle speed limits	NRMM checks all three; we have no stem or roughness data	Land-cover score stands in; say so
Rainfall source changes in 2025	IMD to ERA5, with a different day definition	Documented

10. Status and what is needed from you

Done

- Vehicle table (1,145 vehicles), terrain grid, roads, forecaster, margins, planner, backtest, trade-off sweep, convoy experiment, edge numbers.
- Passes per track, NRMM soil speed, routing by travel time and its backtest; all in the edge bundle too.
- A0 poster: `poster/poster.pdf`. 2-page abstract: `paper/abstract.pdf`.
- Both checked: the abstract is exactly 2 pages, ACM format, with all fonts embedded.

Needed from you before 15 October

1. **Author names, affiliations and emails**, for both files. Review is single-blind, so they must be on the abstract.
2. **Institution logo**, a **project URL** (for the poster's QR code) and a **contact email**.
3. **A read-through by co-authors or your advisor.** The modelling choices listed in Section 9 should be agreed by everyone whose name goes on it.
4. **Check whether your institution must approve defence-related publications.** All data is public, but some institutions require sign-off.
5. **Submit** `paper/abstract.pdf` on EasyChair, track "ICDCN 2027 — Poster".

After acceptance (7 November)

1. Camera-ready and registration by 14 November. The abstract needs ACM's rights text added back; page 2 has room for it.
2. Check the ACM open-access fee for your institution.
3. Print the poster at A0. Proof it at A4 first.
4. One author presents in person, 4–7 January 2027.

11. Where everything is

Folder	Contents
`data/`	Weather record, vehicle file, Earth Engine GeoTIFFs, OpenStreetMap roads
`traffic/`	All code. `paths.py` holds every file location
`outputs/`	Results: `forecaster/`, `terrain/`, `planner/`, `vehicles/`; `forecasting/` is the outdated September comparison
`poster/`	`poster.pdf` (A0), its figures and fonts
`paper/`	The abstract: edit `abstract_template.tex`; the folder uploads to Overleaf as it is
`docs/`	This explainer and its template, the August report, the references; older versions in `archive/`

Rebuilding after a change

Command	What it does
`python -m traffic.check`	Checks packages, files and data loading
`python -m traffic.forecast`	Retrains the forecaster and margins (under a minute)
`python -m traffic.forecast --decisions-only`	Re-tunes the margins after a change to the vehicle table, without retraining
`python -m traffic.mobility`	Passes and level-ground speed of each band's planning vehicle
`python -m traffic.planner`	Backtest and convoys (about 12 minutes); `--tradeoff` for the margin sweep (about 15); `--speed` for the travel-time backtest
`python -m traffic.poster_figs` then `poster_build`	Rebuilds the poster
`python -m traffic.abstract_build`	Rebuilds the abstract and reports its page count
`python -m traffic.explainer_build`	Rebuilds this document

A. Glossary

Term	Meaning
A* / Dijkstra	Algorithms that find the cheapest route on a grid exactly
AHP	Analytic Hierarchy Process: setting weights by having experts compare factors two at a time
Backtest	Testing a method on past data as if live, then checking against what happened
Band (A–E)	A range of one-pass vehicle cone index; results are reported per band
Cone index / RCI	Ground firmness from pushing a cone into the soil; RCI also allows for traffic weakening it
Contact pressure factor (CPF)	Vehicle weight over the area of its tracks or tyres, in psi; NRMM's soil relations differ below and above 4 psi
Drawbar pull (D/W)	The pull a vehicle has left after overcoming the soil's drag, as a fraction of its weight
Edge computing	Running software on a device in the field instead of a data centre
ERA5 / ERA5-Land	Global reconstructions of past weather and land conditions
Fuzzy AND	Combining 0–1 scores by taking the smallest
GEE	Google Earth Engine, a platform for processing satellite data
GRU	Gated recurrent unit: a neural network that reads a sequence of days one at a time [6]
Held back	A planner found no route although one existed: the costly error
Excess firmness (x)	RCI − VCI_1: how far the ground is above what the vehicle needs
Mobility Index (MI)	A US Army formula turning vehicle specifications into one number, from which VCI follows
Motion resistance (R/W)	The soil's drag on a moving vehicle, as a fraction of its weight
NRMM	NATO Reference Mobility Model: the standard model of cross-country vehicle speed
Passes	How many vehicles one track carries before it fails: 1 at VCI_1, 50 at VCI_{50}
NDVI	A satellite index of green vegetation; used here for canopy density
Persistence	The forecast "tomorrow equals today"
Precision / recall	Of warnings given, how many were right / of real events, how many were warned about
Ridge regression	A straight-line model protected against over-fitting
Safety margin	An amount subtracted from the forecast so borderline days count as NO-GO
Slip	The share of track or wheel movement lost to soil shearing instead of moving the vehicle

Stuck	A planner's route crossed ground that was NO-GO on the travel day: the dangerous error
SWI	Soil wetness index: the two soil layers combined, 0 driest to 1 wettest
TWI	Topographic wetness index: from the elevation map, high where water collects
VCI_1 / VCI_{50}	Firmness a vehicle needs to pass once / fifty times on the same track

B. Every formula: where it comes from and how it was set

Each formula is marked **published** (taken from a source, with its numbers), **standard** (a textbook method), or **ours** (a choice made in this project). For ours, the last column says how the numbers were chosen. Section 7 shows each one applied to a real trip.

B.1 Weather to ground firmness

Formula	What it does	Source	How it was set
$s = (\theta - 20)/(49 - 20)$, clipped to 0–1	Scales each soil layer's moisture θ (vol%) to 0 driest, 1 wettest	Standard scaling	20 and 49 are the driest and wettest ERA5 values in the 11 years (20.7, 48.85), rounded
$SWI = 0.35 \cdot s_{0-7\ cm} + 0.65 \cdot s_{7-28\ cm}$	One wetness number for the depth a vehicle works in	Ours	Judgement: vehicles sink into roughly the top 30 cm, most of it in the deeper layer. ANAMOB's critical layers, 0–15 cm for one pass and 15–30 cm for fifty [12], would support about 0.47 / 0.53 for one pass
$RCI = 300 \cdot e^{-2.996 \cdot SWI}$	Firmness falls steeply as soil wets, then levels off	Ours, one end published	Two anchors: 15 psi when saturated, from field lanes where a 4×4 bogged down [2]; 300 psi when dry, assumed. 2.996 = ln(300/15) makes the curve pass through both. The published alternative, ln RCI = a + b·ln(moisture) with constants per soil type [3], needs the soil type
$\times (1 - 0.15 \cdot \min(R_{3\ days}/50\ mm, 1))$	Briefly softens the surface after rain	Ours	Assumed: daily soil moisture misses standing water and surface mud; at most 15%, reached at 50 mm in 3 days
$ET_0 = 0.0023 \cdot (T_{mean} + 17.8) \cdot \sqrt{(T_{max} - T_{min})} \cdot R_a \cdot 0.408$	Evaporation demand, a forecaster input	Published [15]	As published; R_a from latitude and day of year
$API_t = k \cdot API_{t-1} + rain_{t-1}$	Rain memory that fades, a forecaster input	Standard [19]	k = 0.85 and 0.95: a short and a long memory

B.2 The forecast and the call

Formula	What it does	Source	How it was set
$RCI_{t+h} = RCI_t + ½ \cdot (\Delta GRU + \Delta ridge)$	Forecasts the change, not the level, then adds today back	Ours	Firmness barely changes day to day, so learning the change works far better (Section 5.1). GRU and ridge, 3 GRUs averaged, chosen on the 2022–23 validation years

m(band, lead) = smallest margin, in 0.25 psi steps, catching ≥ 90% of NO-GO days	Lowers the forecast so borderline days count as NO-GO	Ours	90% recall is the target; pooled over the band's vehicles; tuned on one test year, scored on the other
NO-GO if RCI − m < VCI_1; SLOW-GO if < VCI_{50}; else GO	The call	Labels published [16]; VCI_1 line published [12]; VCI_{50} line ours	NO-GO below VCI_1 is what VCI_1 means. Putting GO at VCI_{50} (fifty vehicles can follow) is our mapping onto the three labels

B.3 Vehicles

Formula	What it does	Source	How it was set
MI = (CPF·WF/(TEF·GF) + BF − CF)·EF·TF (tracked); WLF in place of BF (wheeled)	Turns a vehicle's specifications into one mobility number	Published [12, 13]	As published. CPF contact pressure; WF weight factor (steps at 50, 70, 100 thousand lb tracked; a line per load range wheeled, the range above 31,500 lb from NRMM II [14]); TEF track or tyre width; GF grousers or chains; BF / WLF load per road wheel or axle; CF clearance/10; EF 1.05 below 10 hp per short ton; TF 1.05 for manual gearboxes
VCI_1 = 7.0 + 0.2·MI − 39.2/(MI + 5.6); VCI_{50} = 19.27 + 0.43·MI − 125.79/(MI + 7.08)	Tracked: firmness needed for 1 and 50 passes	Published [12]	As published; reproduces the M1 example [3]
VCI_1 = (11.48 + 0.2·MI − 39.2/(MI + 3.74))·DCF, or $4.1 \cdot MI^{0.446}$·DCF above MI 115; VCI_{50} = (28.23 + 0.43·MI − 92.67/(MI + 3.67))·DCF	Wheeled: the same	Published [12, 14, 2]	As published. DCF = $(0.15/(\delta/h))^{0.25}$ corrects for tyre deflection; we assume 15% deflection, so DCF = 1. Reproduces six published 8×8 vehicles [14]
Track width, length, road wheels and clearance by weight	Fills in running gear the file lacks	Ours	Straight-line interpolation on weight between six real vehicles, M22 Locust to Tiger II
Tyre size by axle load	The same for wheeled vehicles	Ours	Five standard military tyre sizes, from HMMWV to heavy launcher, picked by the load per axle
Half-track = worse of track unit and front axle	Rates a vehicle with both	Ours	No published formula. Track unit rated as tracked; steered axle rated as wheeled carrying 25% of the weight
Bands A–E at VCI_1 18, 21, 24, 30 psi	Groups vehicles with similar needs	Ours	Read off the fleet's VCI_1 spread so each band is mostly one kind of vehicle
Planning vehicle = band medians	One vehicle stands for each band	Ours	Median VCI_1 and VCI_{50} of the band; majority type; median power-to-weight and contact pressure of that type

B.4 The map

Formula	What it does	Source	How it was set
Slope score	0–1 passability by slope, per vehicle type	Published [1], smoothed by us	Liu et al.'s Table 1, itself vehicle speed on each slope class relative to flat ground. A smooth curve through the middle of each class replaces the steps
Land-cover score	0–1 passability by land cover	Published [1], tree split ours	Liu et al.'s Table 3. Tree cover goes to their forest, bush-wood or grassland row by peak NDVI: 0.75 and above, 0.60–0.75, below 0.60
$TWI = \ln(a / \tan \beta)$	High where water collects	Standard [7, 17]	a is the area draining through a cell (D8 flow routing), β its slope
$shift = 0.10 \cdot clip(z_{TWI}, -2, 2)$; $RCI_{cell} = RCI \cdot e^{-2.996 \cdot shift}$	Spreads one plateau value over the map	Ours	A rule of thumb with no field data: up to 0.2 wetter in hollows, 0.2 drier on ridges, average unchanged. The second formula follows from the firmness formula
Soil score: 0 below VCI_1, then $0.25 + 0.75 \cdot (RCI - VCI_1)/(VCI_{50} - VCI_1)$, up to 1	Soil passability	Ours	0.25 at VCI_1 makes the worst SLOW-GO ground four times slower than firm ground; straight line up to VCI_{50}
$\mu = \min(slope, cover, soil)$	One passability score per cell	Standard fuzzy AND [8]	Compared against Liu's weighted mean in Step 6

B.5 Routes

Formula	What it does	Source	How it was set
$cost = step \cdot \frac{1}{2}(1/\mu_u + 1/\mu_v)$	Relative travel time of a step between cells u and v	Ours	Treats μ as a fraction of full speed, as Liu et al.'s slope scores are
Dijkstra's algorithm	Finds the cheapest route exactly	Standard [18]	—
Firmness needed on pass n = $VCI_1 + (VCI_{50} - VCI_1) \cdot \ln n / \ln 50$	Convoy wear	Ours, ends published	Joins the two published points, 1 and 50 passes [12], with a log curve
Track updates: 4 bytes per cell; forecast: 4 + 7 × 2 = 18 bytes	What crosses the radio link	Ours	A 32-bit cell number per cell of track; a 32-bit day number and seven 16-bit firmness values

B.6 Passes and speed

Formula	What it does	Source	How it was set
$n = 50^{(RCI - VCI_1)/(VCI_{50} - VCI_1)}$	Vehicles one track carries	Ours, ends published	The wear rule above, solved for n

D/W, R/W, TFOR/W, slip S (tables in Step 8)	Pull, drag, most traction and slip on fine-grained soil, from excess firmness	Published [13]	As published (NRMM I subroutines IV3 and IV5), from Rula and Nuttall [12] revised by Turnage (1972); two sets each, for contact pressure below and above 4 psi. Slip is set to 0 rather than NRMM's 1 when the force goes negative downhill
load = sin θ + R/W·cos θ; stuck if TFOR/W·cos θ < load	Force needed on a slope; whether the soil can supply it	Published [13]	As published (subroutine IV8); θ from the elevation difference along each step
v = η·P/(W·load)·(1 − S), capped at top speed	Speed at which engine power meets the force needed	Method published [13]; constant power ours	NRMM builds the pull at each speed from engine torque and gears; with no gearbox data we take constant power. η = 80% assumed. Top speed is the game template value
Paved road: R/W = 0.045 (tracks), 0.007 + 0.0939/psi (tyres)	Rolling resistance on hard surface	Published [13]	Tyre pressure 75 psi, inside the 50–105 psi road pressures of the 8×8s in [14]; road traction 0.8, which only limits steep grades
Off-road speed × land-cover score	Slows forest and shrub	Ours	Stands in for NRMM's vegetation limits, for which we have no stem data
time = Σ step·½(1/v_u + 1/v_v)	Trip time in hours	Ours	Half of each step at each end cell's speed, as for the cost above

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